



Indian Institute of Technology Madras

Voltage Amplifier Circuit Design

(0–1 V Input to 0–50 V Output)

Submitted by
Arpit Katiyar

Contents

1	Voltage Amplifier Circuit (0–1 V to 0–50 V)	2
1.1	Objective	2
1.2	Theory of Operation	2
1.3	Observation with two different OpAms	2
1.4	Circuit Design	2
1.5	Design Calculations	3
1.6	DC Transfer Characteristics	4
1.7	Transient Response with Capacitive Load	5
1.8	Stability Analysis	6
1.9	Noise Analysis	7
1.10	Results Summary	8
1.11	Conclusion	8

1 Voltage Amplifier Circuit (0–1 V to 0–50 V)

1.1 Objective

The objective of this task is to design and simulate a voltage amplifier circuit that converts a low-level control signal in the range of 0–1 V into a high-voltage output of up to 50 V DC. Such amplifiers are commonly used to drive actuators like piezoelectric transducers in optical and precision control systems.

1.2 Theory of Operation

A voltage amplifier increases the amplitude of an input voltage signal by a fixed gain factor. In this design, a high-voltage operational amplifier (or an idealised amplifier model in LTspice) is configured in a non-inverting topology to achieve a voltage gain of approximately 50.

For a non-inverting op-amp configuration, the closed-loop voltage gain is given by:

$$A_v = 1 + \frac{R_f}{R_g} \quad (1)$$

where R_f is the feedback resistor and R_g is the resistor connected to ground.

The amplifier must maintain stability while driving a capacitive load that models a piezoelectric actuator (typically 10–200 nF).

1.3 Observation with two different OpAms

Two different operational amplifier models were used for the LTspice simulations: the UniversalOpAmp2 and the high-voltage op-amp LTC6090.

All the assigned tasks were simulated using both models. It was observed that the LTC6090, being a specific high-voltage op-amp, exhibits output limitations imposed by its supply voltage rails. In contrast, the UniversalOpAmp2 produced the expected output even when the supply rails were varied, indicating that its behavior is largely independent of the supply voltage rails.

The results of the simulations for both op-amp configurations corresponding to the given tasks are attached below.

1.4 Circuit Design

The voltage amplifier circuit consists of:

- High-voltage operational amplifier
- Feedback resistor network to set gain

- Capacitive load representing a PZT actuator
- Decoupling and biasing components

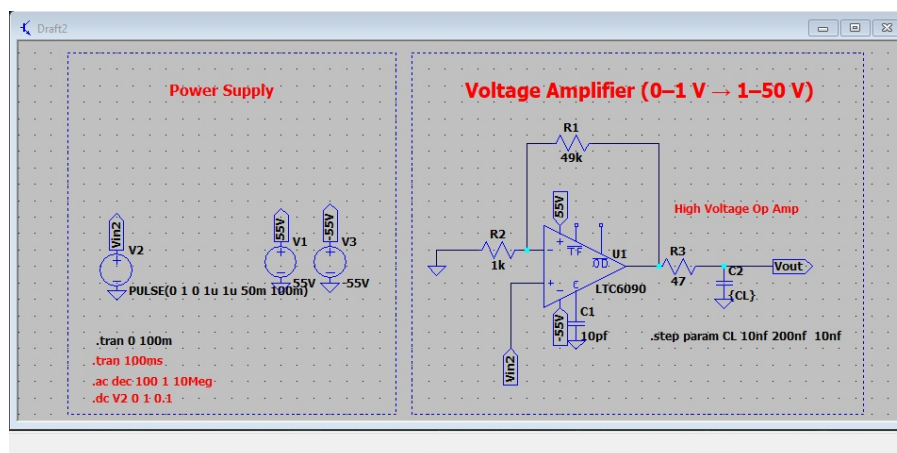


Figure 1: LTspice schematic of the voltage amplifier circuit with LTC6090

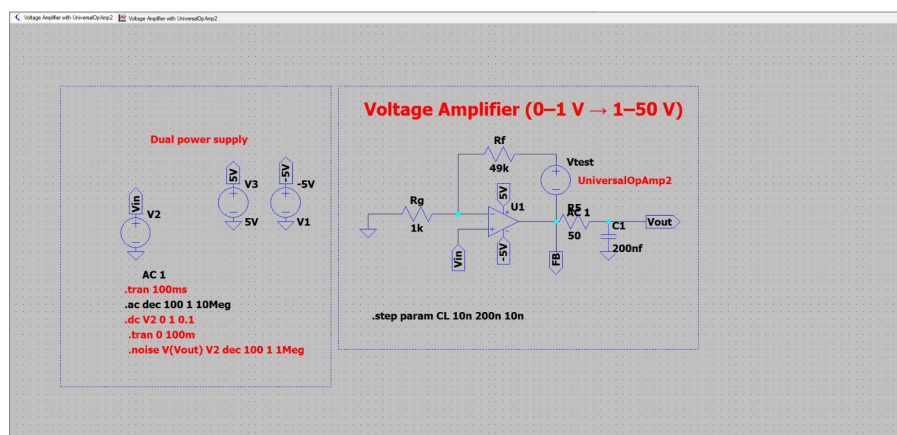


Figure 2: LTspice schematic of the voltage amplifier circuit with UniversalOpAmp2

1.5 Design Calculations

To achieve an output of 50 V for a 1 V input:

$$A_v = 50$$

Using the gain equation:

$$50 = 1 + \frac{R_f}{R_g} \Rightarrow \frac{R_f}{R_g} = 49$$

Choosing:

$$R_g = 1 \text{ k}\Omega, \quad R_f = 49 \text{ k}\Omega$$

This resistor combination provides the required voltage gain.

1.6 DC Transfer Characteristics

DC sweep analysis is performed to verify linear operation over the full input range.

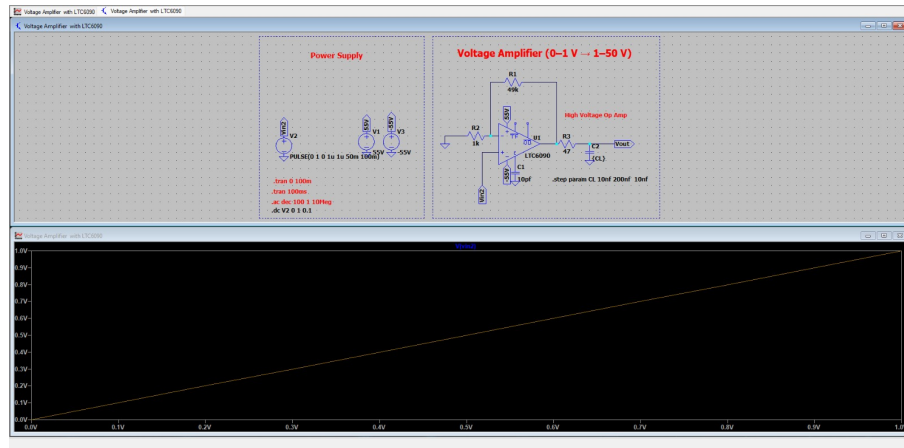


Figure 3: DC transfer curve: input voltage vs input voltage with LTC6090

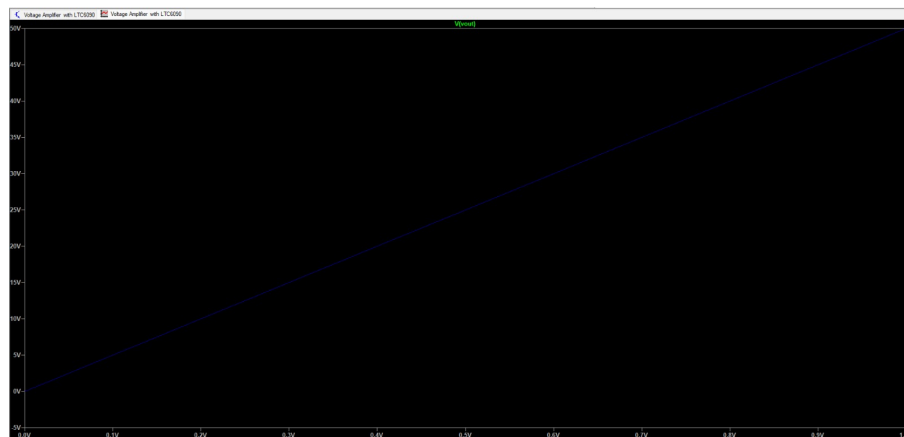


Figure 4: DC transfer curve: input voltage vs output voltage with LTC6090

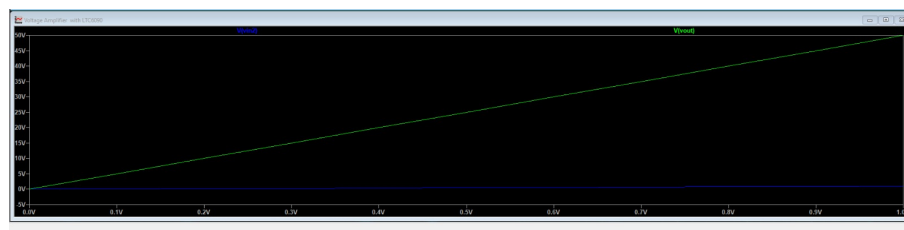
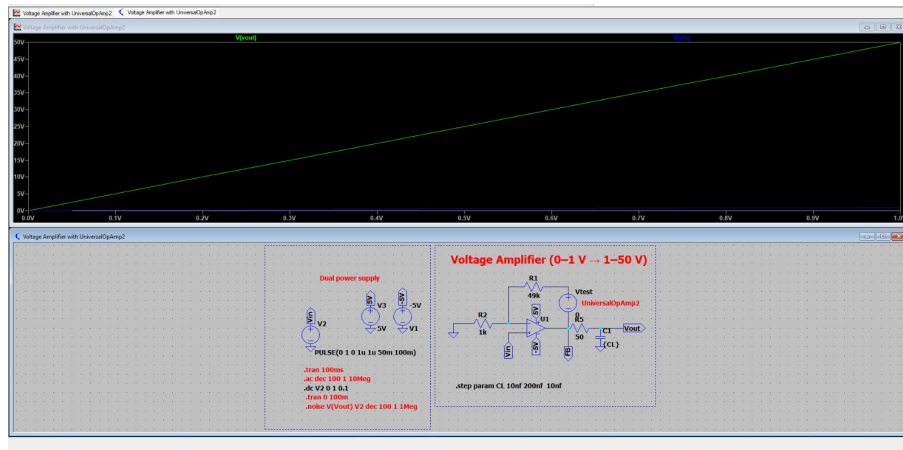
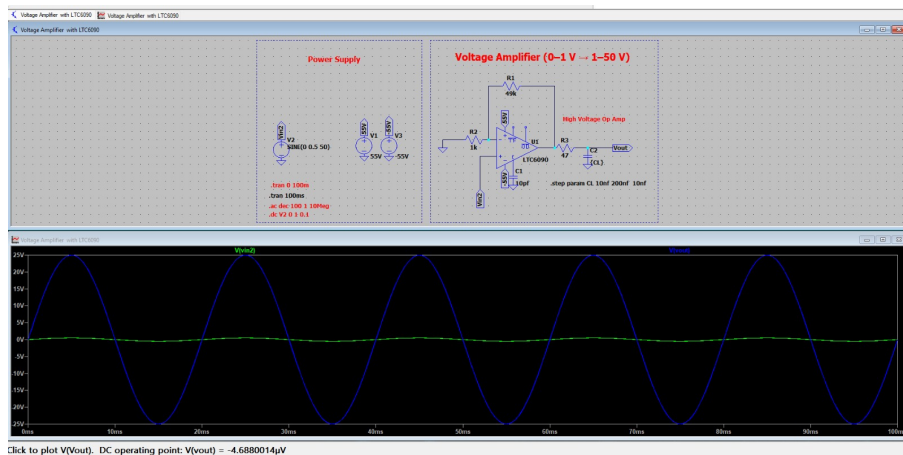


Figure 5: DC transfer curve: input voltage vs output voltage and input voltage vs input voltage with LTC6090



1.7 Transient Response with Capacitive Load

A capacitive load ranging from 10 nF to 200 nF is added to simulate a piezoelectric actuator. Transient analysis is performed to observe rise time, overshoot, and settling behaviour.



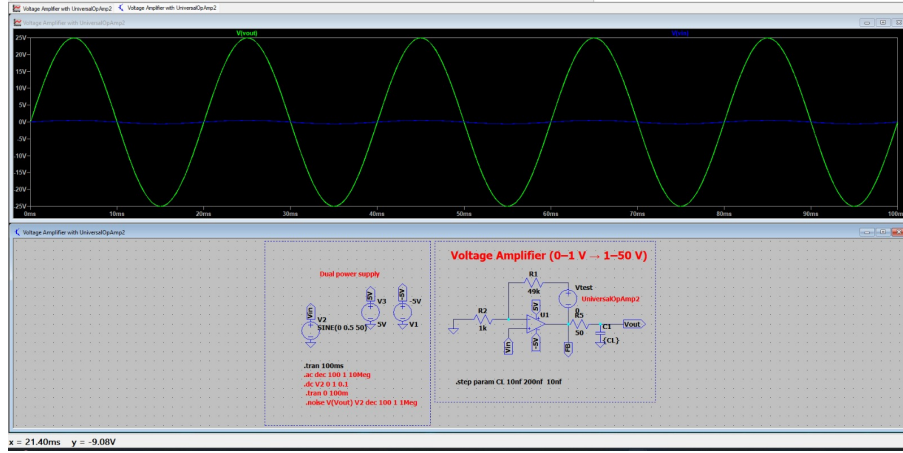


Figure 8: Transient response with capacitive load with UniversalOpAmp2

1.8 Stability Analysis

AC analysis is used to determine the loop stability of the amplifier. The phase margin is evaluated from the Bode plot to ensure stable operation. Closed-loop stability was verified using AC (Bode), transient, and noise analyses. The Bode plot shows a smooth magnitude roll-off with no gain peaking and a gradual phase decrease, indicating stable closed-loop operation. Step and load transient simulations exhibit damped ringing with finite settling time for capacitive loads up to 200 nF, and no sustained oscillations were observed. Noise analysis further confirms stability by showing no bandwidth-related noise peaking.

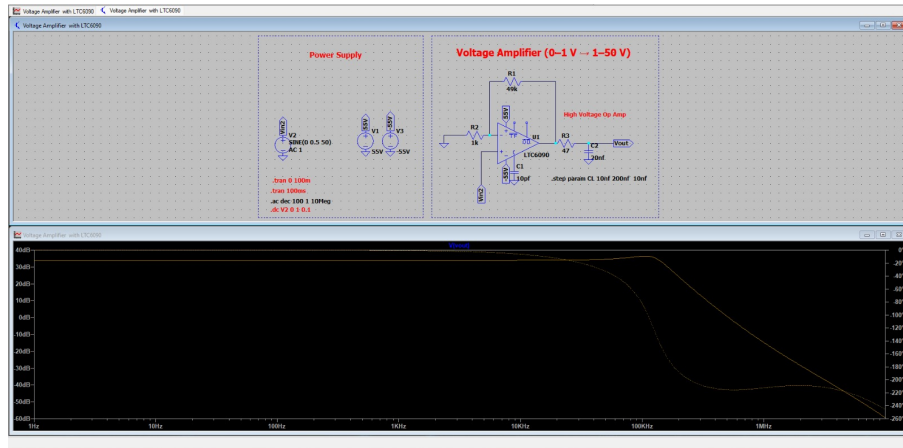


Figure 9: Bode plot showing gain and phase margin with LTC6090

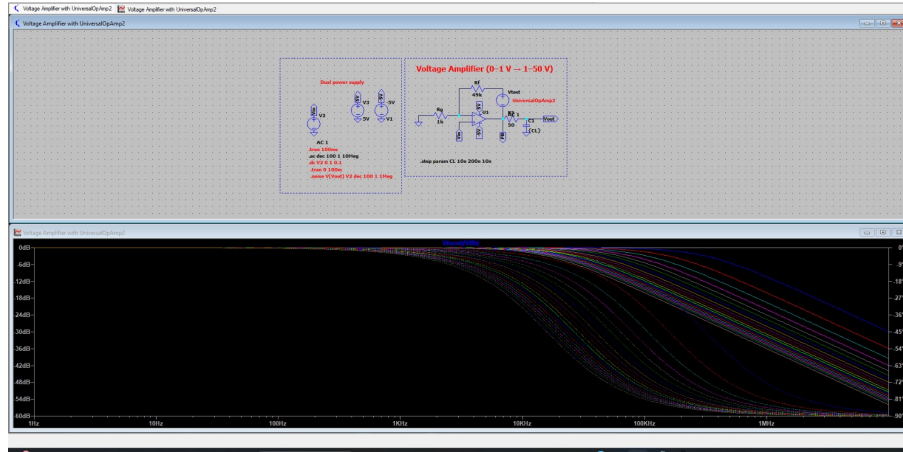


Figure 10: Bode plot showing gain and phase margin with UniversalOpAmp2

1.9 Noise Analysis

Output noise analysis is performed to estimate the noise performance of the voltage amplifier under steady-state conditions.

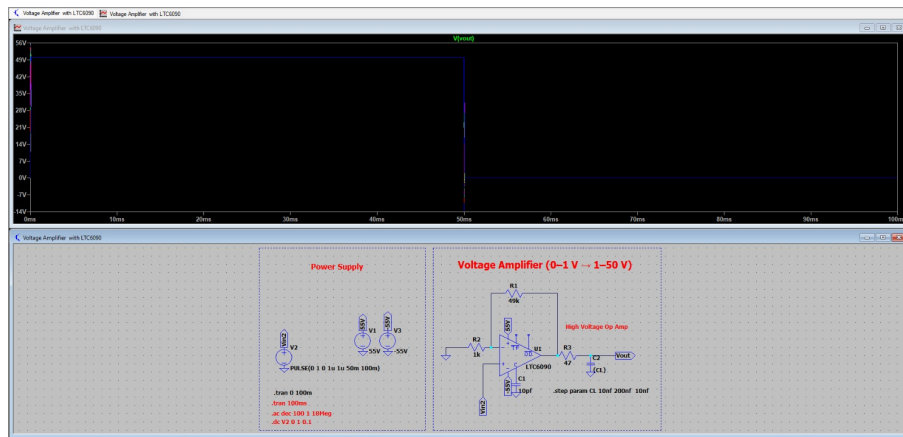


Figure 11: Output noise spectrum of the voltage amplifier with LTC6090

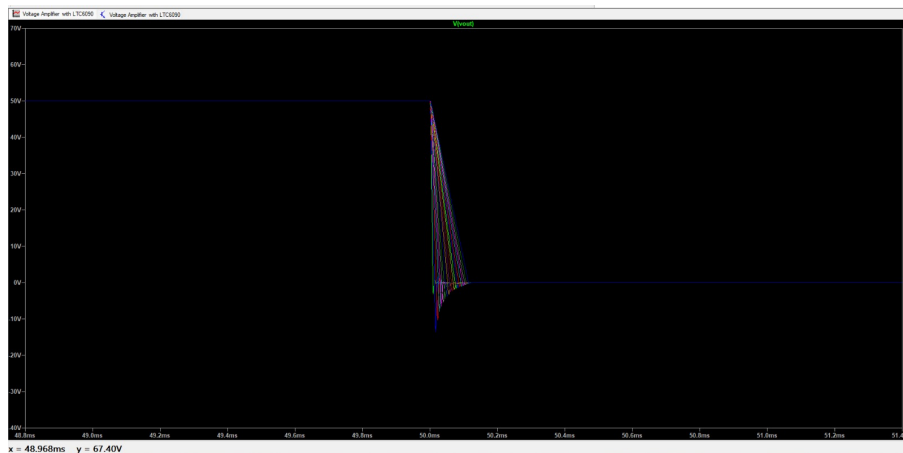


Figure 12: Output noise spectrum of the voltage amplifier LTC6090

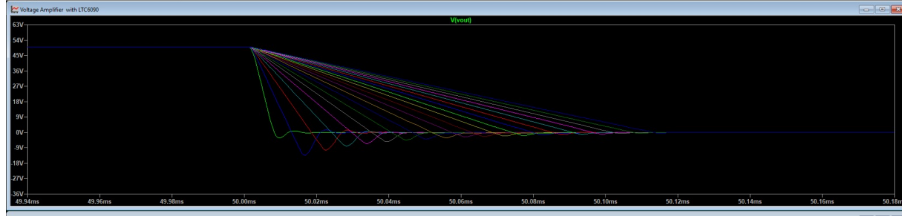


Figure 13: Output noise spectrum of the voltage amplifier with LTC6090

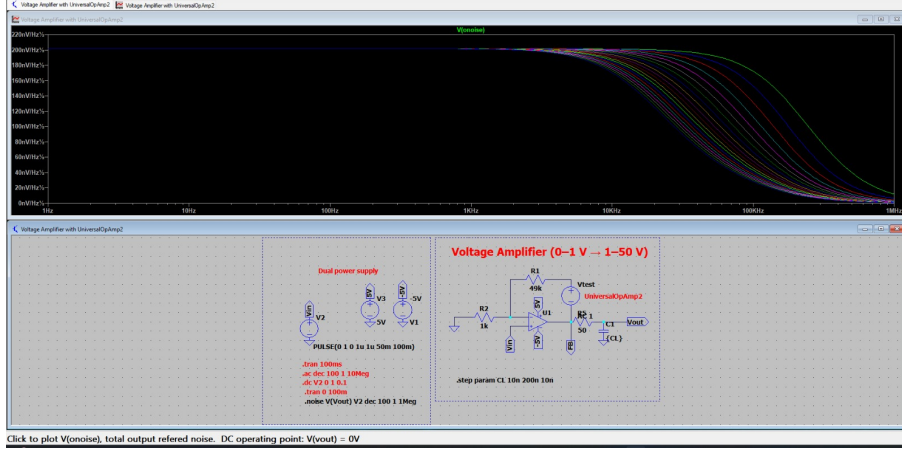


Figure 14: Output noise spectrum of the voltage amplifier with UniversalOpAmp2

1.10 Results Summary

Table 1: Voltage Amplifier Performance Summary

Parameter	Value
Voltage Gain	~ 50
Maximum Output Voltage	50 V
Load Capacitance	10–200 nF
Phase Margin	$> 60^\circ$
Settling Time	To be measured

1.11 Conclusion

The voltage amplifier successfully meets the design requirements of converting a 0–1 V input signal into a high-voltage 0–50 V output while maintaining stability under capacitive loading. The simulation results demonstrate suitability for driving piezoelectric actuators in precision control applications.